UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

Project Report

Course Title: Data Structures

Course Code: (CS–201 L)

Submitted To: Ma'am Muniba Khan

Project Title: Intelligent Route Engine

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1. Introduction

The Intelligent Route Engine is a C++ based project developed to find optimal routes between major cities of Pakistan using different graph search algorithms. The system models cities as nodes and roads as weighted edges, allowing users to select a source city, destination city, and a preferred routing strategy.
This project demonstrates the practical implementation of classical and informed search algorithms studied in Data Structures and Algorithms (DSA).

2. Features

The main features of the system are:

- Supports **40 major cities of Pakistan**
- Graph-based representation of cities and routes
- Multiple route-finding algorithms available
- Displays route path, distance, and cost
- Console-based user interaction
- Efficient handling of different routing preferences

3. Objective of the Project

The objective of this project is to:

- Implement and compare different **search algorithms**
- Understand real-world applications of **graphs**
- Provide an efficient route-finding system
- Analyze performance differences between uninformed and informed searches
- Strengthen understanding of DSA concepts through implementation

4. Project Scope

This system can be used in:

- Navigation and route planning systems
- Logistics and transportation planning
- Educational purposes (DSA learning tool)
- Simulation of intelligent transportation systems
- Decision support systems for shortest or cheapest routes

5. Tools and Techniques

The following tools and techniques were used:

- **Programming Language:** C++
- **Data Structures:** Graphs, Arrays, Queues

- **Algorithms Used:**
  - Breadth First Search (BFS)
  - Depth First Search (DFS)
  - Uniform Cost Search (UCS)
  - Dijkstra's Algorithm
  - A* Search Algorithm
- **Development Environment:** Dev-C++ / Code::Blocks
- **Mathematical Technique:** Haversine Formula (for heuristic calculation)

6. Implementation

The project is implemented using an adjacency list graph structure where:

- Cities are represented as nodes
- Routes are represented as weighted edges
- Each algorithm is implemented as a separate function

6.1 Code Implementation

The system includes the following major functions:

- buildGraph() – Constructs the city graph
- BFS() – Finds minimum stop routes
- DFS() – Traverses graph paths
- UCS() – Finds minimum cost routes
- Dijkstra() – Finds shortest distance routes
- AStar() – Finds smart and fast routes using heuristics

6.2 Screenshots and Figures

Figure 1: List of available cities displayed to the user:

```
======================================================================
  INTELLIGENT ROUTE ENGINE - 38 PAKISTANI CITIES
======================================================================

  Available Cities (Total: 40):
  ------------------------------------------------------------
  Punjab (0-14):
  0. Lahore
  1. Faisalabad
  2. Rawalpindi
  3. Multan
  4. Gujranwala
  5. Sialkot
  6. Bahawalpur
  7. Sargodha
  8. Sheikhupura
  9. Jhang
  10. Rahim Yar Khan
  11. Kasur
  12. Okara
  13. Mianwali
  14. Hafizabad

  Sindh (15-22):
  15. Karachi
  16. Hyderabad
  17. Sukkur
  18. Larkana
  19. Nawabshah
  20. Mirpur Khas
  21. Thatta
  22. Jacobabad

  KPK (23-30):
  23. Peshawar
  24. Mardan
  25. Abbottabad
  26. Mingora
  27. Nowshera
  28. Charsadda
```

```
29. Mansehra
30. Swabi

Balochistan (31-36):
31. Quetta
32. Gwadar
33. Khuzdar
34. Turbat
35. Sibi
36. Zhob

Federal/Others (37-38):
37. Islamabad
38. Muzaffarabad
39. Gilgit
----------------------------------------------------------------
```

Figure 2: Menu showing different route preferences:

```
 Enter source city index (0-39): 15
 Enter destination city index (0-39): 27

+- ROUTE PREFERENCE OPTIONS -+

1. Minimum number of stops (BFS)
2. Traversal path (DFS)
3. Cheapest route (UCS)
4. Shortest distance (Dijkstra)
5. Smart & fast route (A* Search)

Enter your choice (1-5): |
```

Figure 3: Output showing minimum stop route using BFS:

```
 Enter your choice (1-5): 1

----------------------------------------------------------------------
 ==> You selected: Minimum Stops Route (BFS)
----------------------------------------------------------------------

 ==> BFS Path (Minimum Stops):
 ------------------------------------------------------------
 Route: Karachi -> Quetta -> Zhob -> Peshawar -> Nowshera
 Total Stops: 4
 Total Cities Visited: 5
 ------------------------------------------------------------
----------------------------------------------------------------------
```

Figure 4: Smart and fast route generated using A* algorithm.:

```
 5. Smart & fast route (A* Search)

 Enter your choice (1-5): 5

-----------------------------------------------------------------------
 ==> You selected: Smart & Fast Route (A* Search)
-----------------------------------------------------------------------

 ==> A* Path (Smart & Fast):
 ------------------------------------------------------------
 Route: Karachi -> Khuzdar -> Quetta -> Zhob -> Peshawar -> Mingora
 Total Distance: 1370 km
 Heuristic: Straight-line distance based on coordinates
 ------------------------------------------------------------
-----------------------------------------------------------------------
```

Figure 5: Final route output when program ends:

```
----------------------------------------------------------------------

  Want to find another route? (Y/N): n


======================================================================
  THANK YOU FOR USING THE SYSTEM
======================================================================
  +-- Total routes searched: 2  --+
  Goodbye! Safe travels!
======================================================================
```

7. Results

The system successfully calculates routes between selected cities using different algorithms.

Sample Output:

- **Algorithm Used:** A* Search
- **Route:** Lahore → Faisalabad → Jhang → Multan → Rahim Yar Khan → Karachi
- **Total Distance:** 1185 km

The results clearly show how different algorithms produce different paths based on optimization criteria.

8. Conclusion

The Intelligent Route Engine effectively demonstrates the application of graph search algorithms in real-world route planning. The project highlights the strengths and weaknesses of each algorithm and shows how informed search (A*) performs better in terms of efficiency and optimality.
This project strengthened our understanding of graphs, search techniques, and algorithm design.

9. Future Works

Possible future enhancements include:

- Graphical User Interface (GUI)
- Real-time traffic handling
- Integration with maps
- Dynamic input using files or web interface
- Mobile or web-based version
- Optimization using priority queues

10. References

1. **Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C.** (2009).
   Introduction to Algorithms (3rd ed.). Cambridge, MA, USA: MIT Press.
   Available at:
   https://scholar.google.com/scholar?q=Introduction+to+Algorithms+Cormen
2. **Sedgewick, R., & Wayne, K.** (2011).
   Algorithms (4th ed.). Boston, MA, USA: Addison-Wesley.
   Available at: https://scholar.google.com/scholar?q=Algorithms+Sedgewick+Wayne
3. **GeeksforGeeks.**
   Graph Data Structure and Algorithms.
   Available at: https://www.geeksforgeeks.org/graph-data-structure-and-algorithms/
4. **Pakistan City Coordinates (Public Data).**
   Publicly available geographical coordinate datasets used for heuristic calculations in A* search.
5. **Course Study Material.**
   Lecture notes and course resources provided by the instructor for understanding graph algorithms and search techniques.